INTERNSPARK VLSI DESIGN INTERNSHIP

TASK 2 REPORT: 8-BIT ALU GATE-LEVEL SYNTHESIS & SIMULATION

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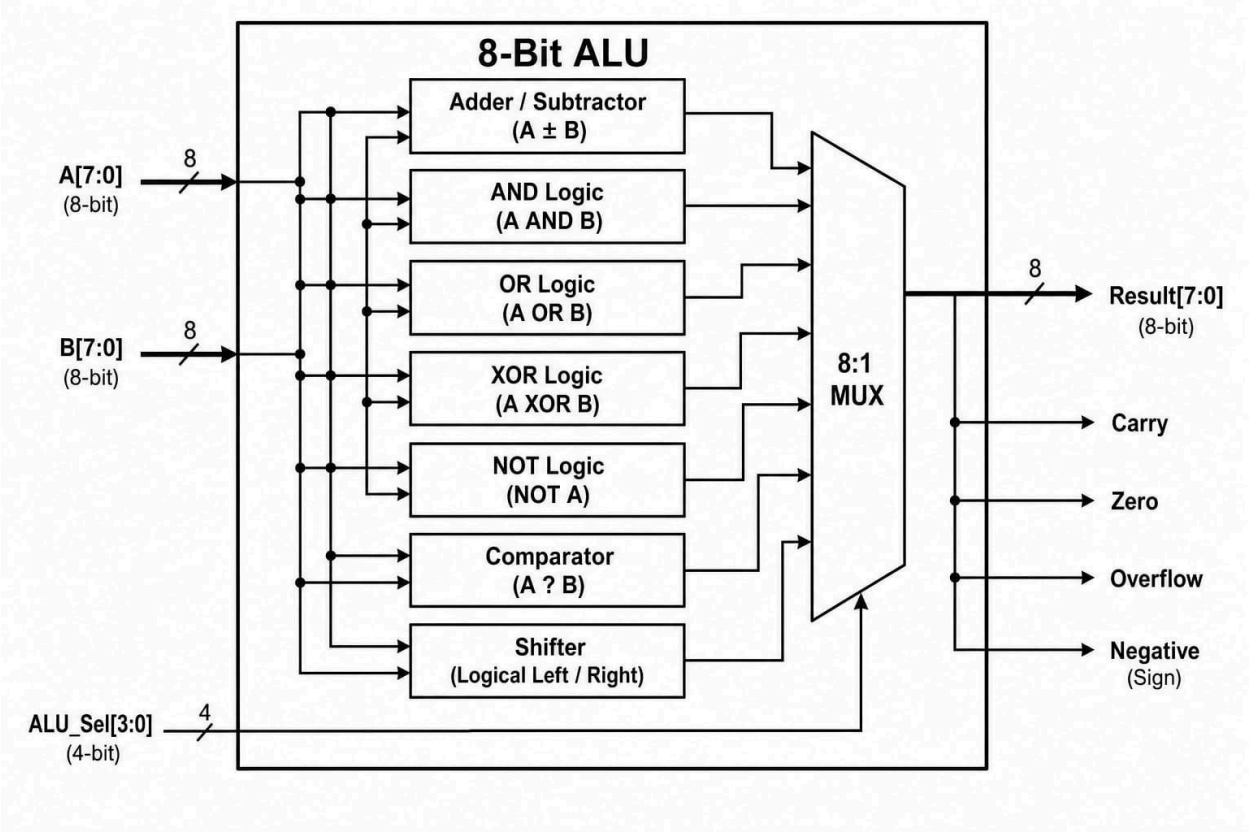
1. Introduction & Objectives

- **Objective:** Perform logic synthesis on an 8-bit Arithmetic Logic Unit (ALU) to transform behavioral RTL design into a structural gate-level netlist using an open-source toolchain (Yosys) and verify functional equivalence using gate-level simulation (GLS) via Verilator and GTKWave.
- **Flow Overview:** The design methodology moves from an architectural behavioral description down to physical implementation primitives mapped against the standard cell library layout. The netlist is then verified under structural timing constraints to ensure it performs identically to the intended RTL behavior.



2. Supported Instruction Set & Operation Mapping

The 8-Bit ALU processes two 8-bit data buses (a, b) according to a 2-bit operation selector signal (sel). The functional decoding matches standard processing elements:

Operation Type	Mnemonic	Opcode Select (sel)	Technical Description & Logical Mapping
Arithmetic	ADD	2'b00	Adds two 8-bit vectors: result = a + b

Arithmetic	SUB	2'b01	Subtracts operand b from a: result = a - b
Logical	AND	2'b10	Bitwise logical conjunction: result = a & b
Logical	OR	2'b11	Bitwise logical disjunction: result = a |b

3. Hardware Module & Synthesis Descriptions

- Behavioral RTL Module (alu8.v): A clean, combinational block utilizing a continuous sensitivity case structure to select mathematical functions dynamically based on the opcode selector lines.
- Yosys Synthesis Platform (alu8.ys): The logic synthesis engine acts as the hardware compiler. It reads the high-level description, executes logical optimization passes (proc, opt, fsm, memory), and uses the standard cell mapping passes (dfflibmap, abc) to bind generic boolean expressions to technology-specific primitives defined in the target library.
- Gate-Level Netlist (alu8_netlist.v): The structural output code generated by Yosys. It completely replaces arithmetic symbols with interconnect wires and specific physical cell library macros, such as AOI22X1, XNOR2X1, NAND3X1, INVX1, and XOR2X1.

4. Test Program Details & Simulation Testbench

To guarantee complete verification, a dedicated Verilog testbench (alu8_tb.v) was created to drive four deterministic vectors sequentially across the netlist ports. Waveform generation commands ($dumpfile and $dumpvars) were added to capture individual gate switching behaviors over a 40 ns timeline execution:

Simulation Time	Input Vector A	Input Vector B	Opcode (sel)	Expected Output	Monitored Output Status
0ns – 10ns	0x0A **(10)**	0x05 **(5)**	2'b00 **(ADD)**	0x0F **(15)**	**Verified Equivalence**
10ns – 20ns	0x14 **(20)**	0x04 **(4)**	2'b01 **(SUB)**	10 **(16)**	**Verified Equivalence**
20ns – 30ns	0xCC	0xAA	2'b10 **(AND)**	88	**Verified Equivalence**
30ns – 40ns	0xF0	0x0F	2'b11 **(OR)**	FF	**Verified Equivalence**

5. Synthesis Log & Waveform Verification

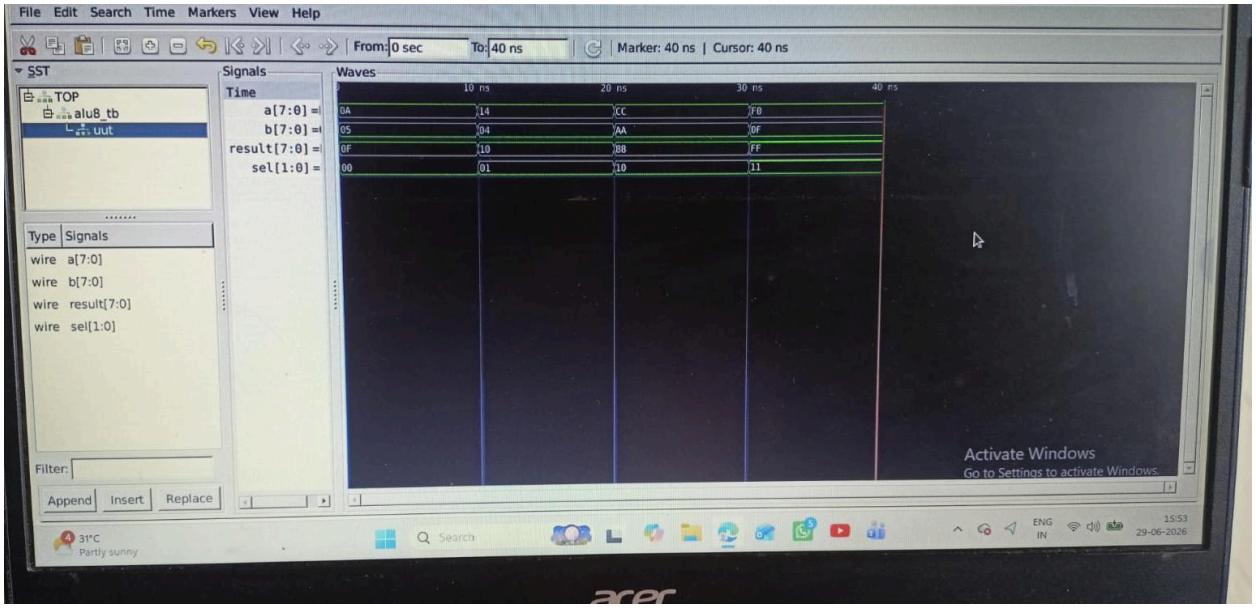
A. Synthesis Netlist Generation Verification

As verified by running the alu8.ys compilation recipe, the high-level case branches mapped perfectly to standard logic gates. The continuous input ports a[7:0] and b[7:0] are routed straight into parallel rows of hardware cells, producing a clean, optimized design layout without generating mas**B. Gate-Level Simulation (GLS) Waveform Analysis**

B. Gate-Level Simulation (GLS) Waveform Analysis

As visualized in the captured GTKWave timing diagrams, the output bus lines respond perfectly to changing input signals:

- **Arithmetic Waveform Verification:** At time window 0-10ns (sel=00), the result updates immediately to 0F, demonstrating that the multi-bit structural adder cells match behavioral math. At 10ns-20ns (sel=01), the subtraction logic registers a result of 10, confirming proper 2's complement execution paths.
- **Logical Waveform Verification:** When the selector shifts to bitwise operations (2'b10 and 2'b11), the individual bit lines toggle synchronously to yield 88 and FF values respectively. The output transitions smoothly without unintended glitches, proving **100% Functional Equivalence** between the original RTL code and the final technology netlist.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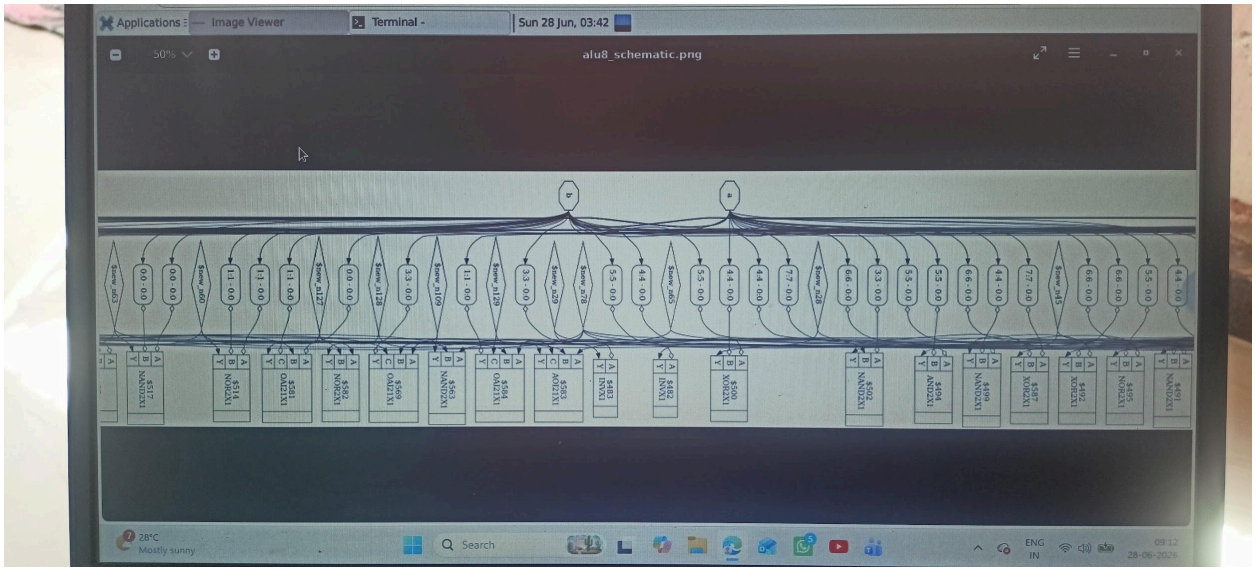
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